Quadratic formula

1 The standard form of a quadratic equation is $ax^2 + bx + c = 0$. Identify the values of a, b and c in the following quadratic equations:

a $8x^2 - 3x = 0$

b $8x^2 + 3 = 0$

c $5x^2 + 6x = 10$

d $3x^2 - 8x + 2 = 9x - 7$

2 Solve the following equations by using the quadratic formula:

a $x^2 + 11x + 28 = 0$

b $4x^2 + 6x + 2 = 0$

c $4x^2 - 7x - 15 = 0$

3 To solve $x^2 - x = 2$, Emma substitutes $a = 1$ and $b = -1$ into the quadratic formula. What value of c should she substitute into the quadratic formula?

4 Consider the quadratic equation $9y^2 = 8y$. Suppose $b = 8$ is used in the quadratic formula:

a State the value of a that should be used.

b State the value of c that should be used.

5 Solve the following equations below using quadratic formula, leaving your answer in radical form.

a $x^2 - 5x - 2 = 0$

b $4x^2 - x - 10 = 0$

c $-2x^2 - 15x - 4 = 0$

6 Solve the following equations below using quadratic formula, leaving your answers as decimals to the nearest thousandth.

a $x^2 + 7x - 3 = 0$

b $5x^2 + 9x + 2 = 0$

7 Solve the equation $1.8x^2 + 5.2x - 2.3 = 0$ using quadratic formula, leaving your answers as decimals to the nearest tenth.

8 Consider the equation $2x^2 = 14$.

a Use the quadratic formula to find the solutions of the equation. Give your answer as an exact value.

b Give the solutions as decimals rounded to the nearest tenth.

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9 Solve the following equation by using the quadratic formula.

a $x\left(x-4\right)=5$

b $x\left(x+9\right)=-20$

Discriminant

10 For a particular quadratic equation, $b^2-4ac=0$. What does this tell us about the solutions of the quadratic?

11 The solutions of a quadratic equation are 9 and -9. What does this tell us about the value of b^2-4ac?

12 For each of the following equations:

i Find the value of the discriminant.

ii State the number of real solutions the equation has.

a $x^2+6x+9=0$

b $4x^2-6x+7=0$

c $2x^2-2x=x-1$

13 State whether the following equations have any real solutions:

a $-x^2-8x-16=0$

b $-2x^2+5x-6=0$

c $-x^2+6x+1=0$

14 Consider the equation $4x^2-4x+1=0$.

a Find the discriminant.

b Now, describe the nature of the roots.

15 Consider the equation $x^2-8x-48=0$.

a Find the discriminant.

b Now, describe the nature of the roots.

c Find the solutions of the equation.

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Quadratics with complex solutions

16 Below is the result after using the quadratic formula to solve a quadratic equation:

$$x = \frac{10 \pm \sqrt{-1}}{8}$$

How many real solutions does the quadratic equation have?

17 Below is the result after using the quadratic formula to solve a quadratic equation:

$$x = \frac{-3 \pm \sqrt{9 - 4 \times 56}}{16}$$

How many real solutions does the quadratic equation have?

18 Solve the following, leaving your answers in the form $a \pm bi$:

a $x^2 + 2x + 5 = 0$

b $25 = x(6 - x)$

c $x^2 + 10x + 28 = 0$

d $x^2 - 4x + 29 = 0$

e $x^2 - 10x + 27 = 0$

f $4x^2 + 20x + 33 = 0$

g $-4x^2 + 4x - 13 = 0$

h $12 - 22x + 5x^2 = 4 - 4x^2 - 10x.$

i $\dfrac{-10x - 40}{x} = x$

j $\dfrac{x^2}{2} = 2x - 18$

k $9 + \dfrac{115}{x^2} = \dfrac{-24}{x}$

19 Find the zeros of the function $8x^2 + 7x + 3 = 0$ algebraically, leaving your answer in the form $a \pm bi$.

20 Consider the quadratic equation $ax^2 + 2x + c = 0$. State whether the following pairs of values for a and c will give non-real solutions:

a $a = -5$ and $c = -6$

b $a = -5$ and $c = 6$

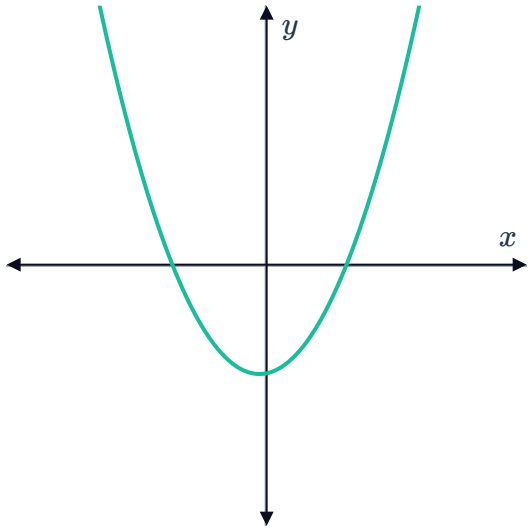
c $a = 5$ and $c = -6$

d $a = 5$ and $c = 6$

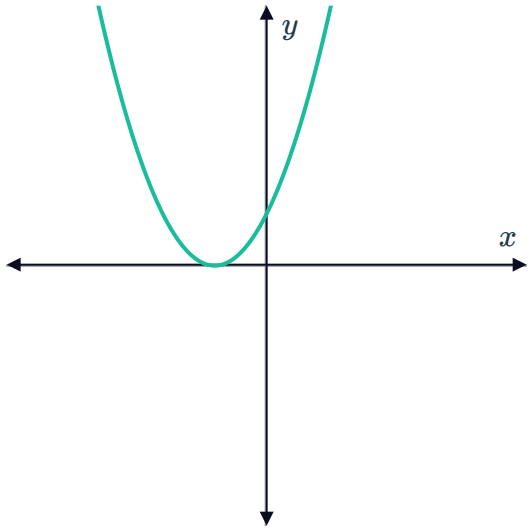
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21 How many non-real solutions do the equations of the following quadratic curves have?

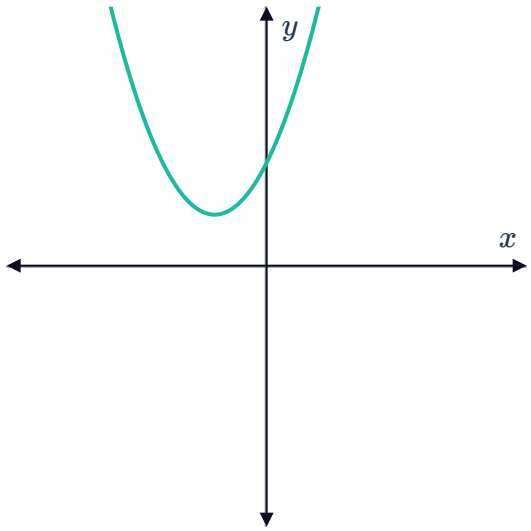
a



b



c



22 Consider the equation in x : $mx^2 + 2x - 1 = 0$. Given that it has two solutions, state the possible values of m.

23 Find the values of n for which $x^2 - 8nx + 1296 = 0$ has one solution.

24 Find the value of k for which $16x^2 + 8x + k = 0$ has equal roots.

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25 Consider the equation $2x^2 + 8x + k = 0$.

a Find the discriminant in terms of k.

b Find the value of k when the equation has:

i Equal solutions

ii Real solutions

iii No real solutions.

iv Real and distinct solutions.

26 Consider the equation $(k+4)\,x^2 + 10x + 3 = 0$.

a Find the discriminant in terms of k.

b Find the value of k when the equation has:

i Equal solutions

ii Real solutions

iii No real solutions

iv Real and distinct solutions

Discriminant and parabolas

27 To find the x-intercepts of a particular parabola, $ax^2 + bx + c = 0$, Katrina used the quadratic formula and found that $b^2 - 4ac = -5$. How many x-intercepts does the parabola have?

28 The graph of $y = m - 9x - 3x^2$ has no x-intercepts.

a Solve for the possible values of m.

b What is the largest possible integer value of m?

29 Solve for the value(s) of k for which $y = 4x^2 - 4x + k - 15$ just touches the x-axis.

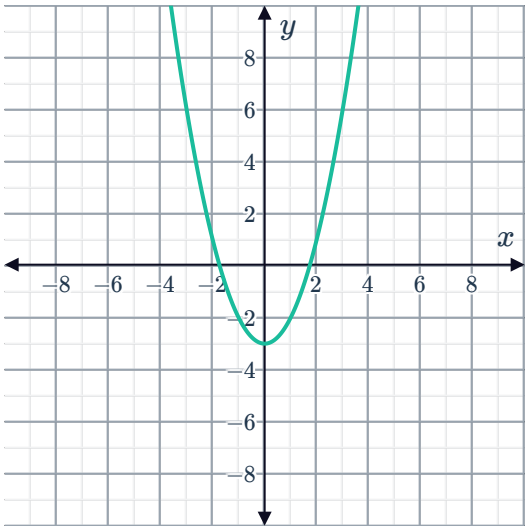
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30 For each of the following graphs of a quadratic $f(x) = ax^2 + bx + c$, with discriminant $b^2 - 4ac$:

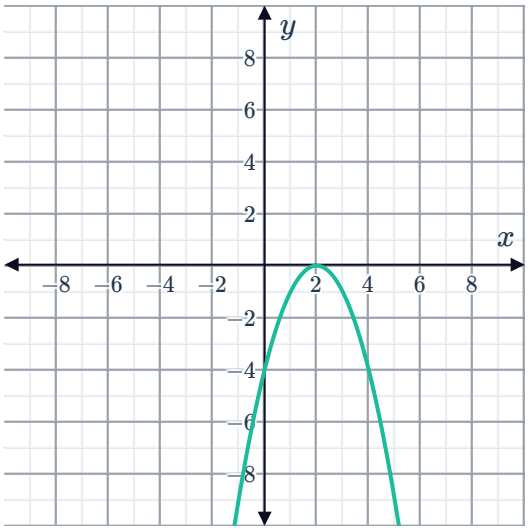
i State whether $a > 0$ or $a < 0$.

ii State whether $b^2 - 4ac > 0$, $b^2 - 4ac < 0$ or $b^2 - 4ac = 0$.

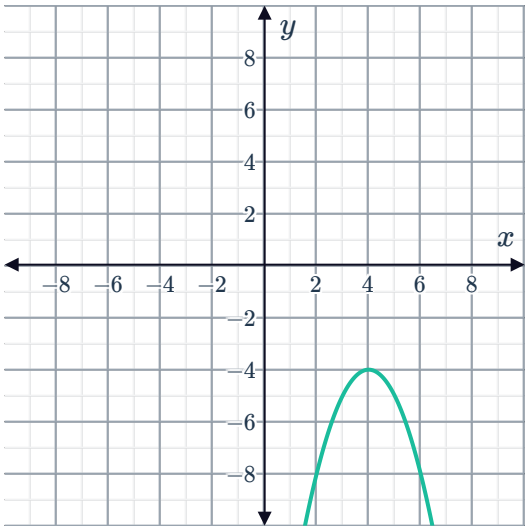
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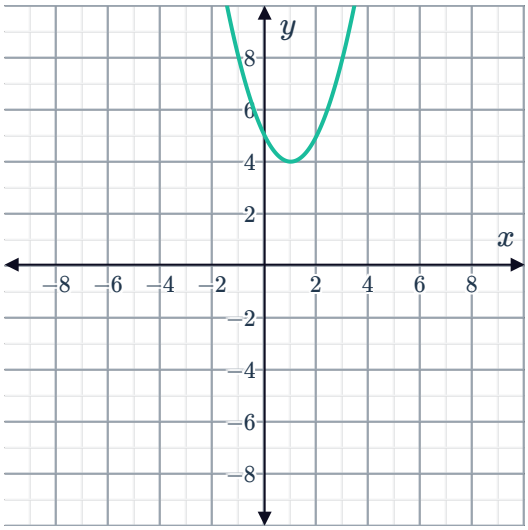
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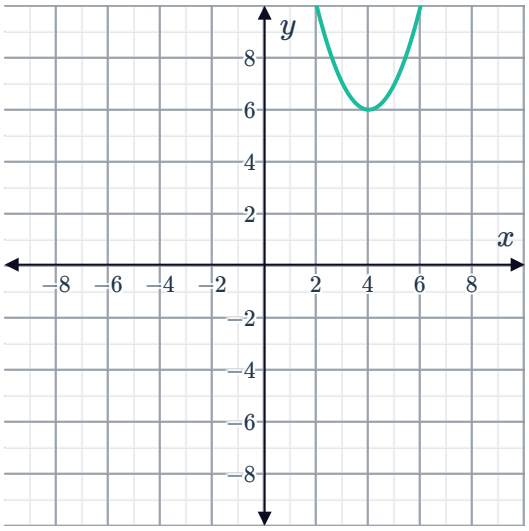


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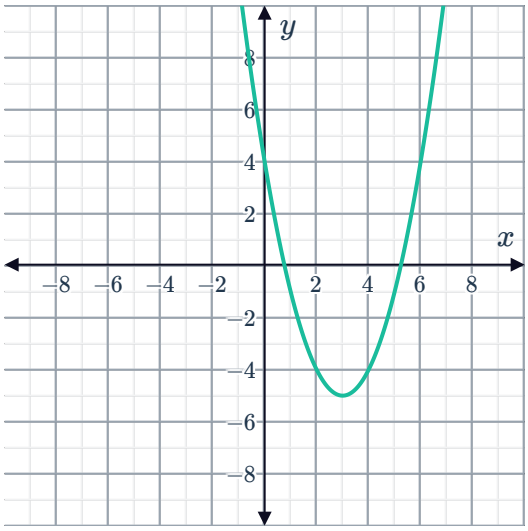


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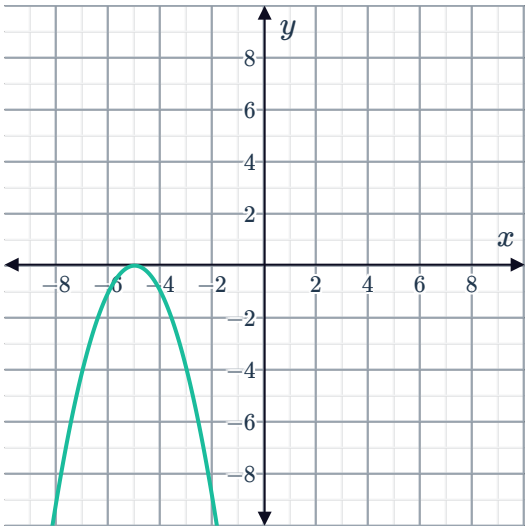
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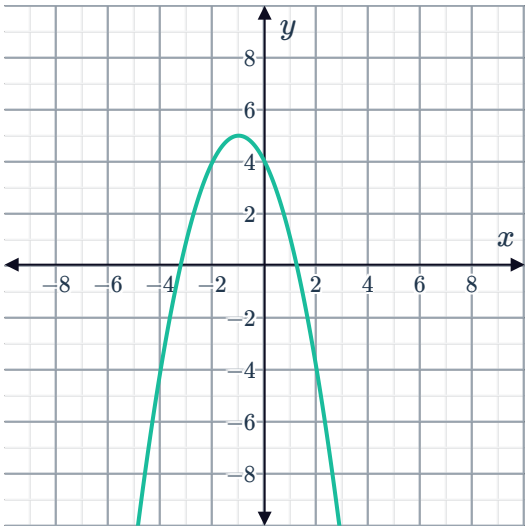
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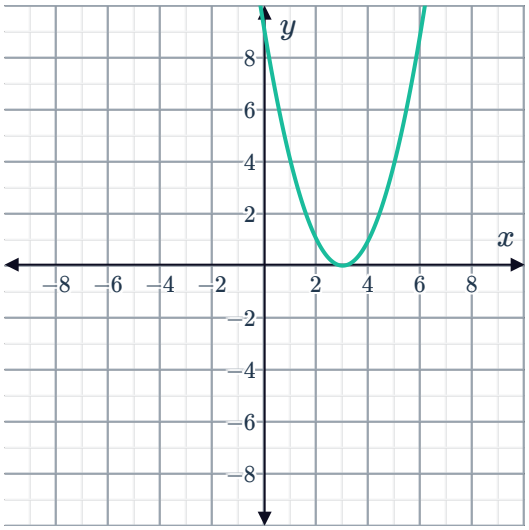
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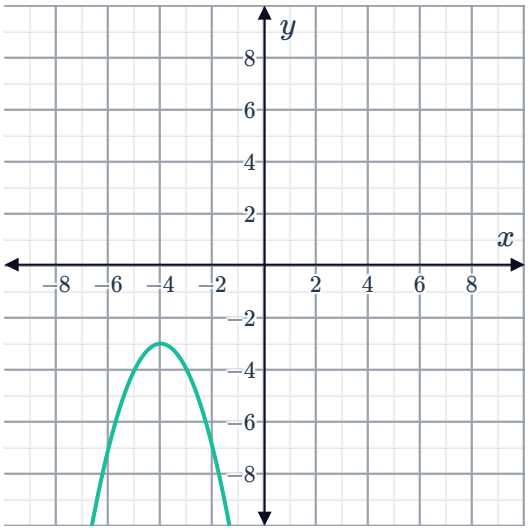
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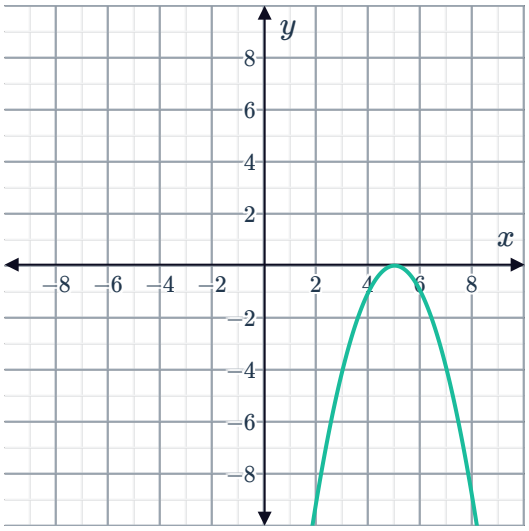


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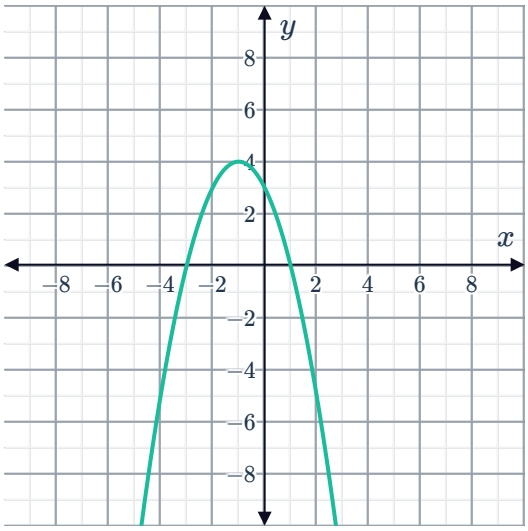


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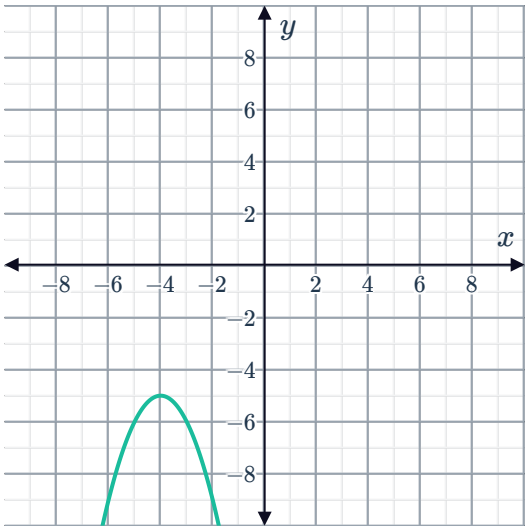
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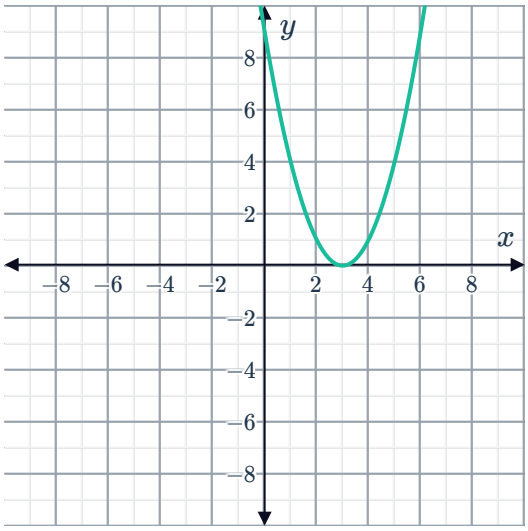
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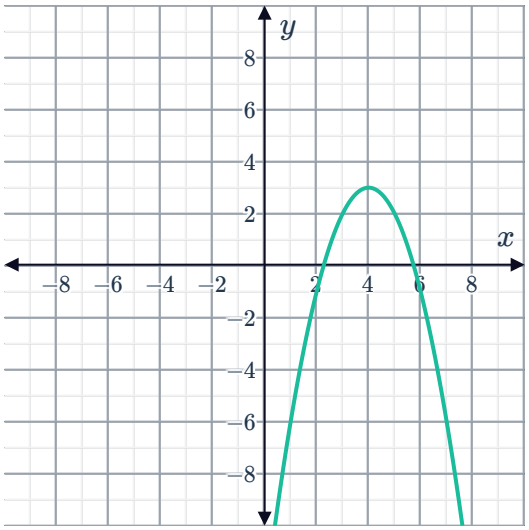
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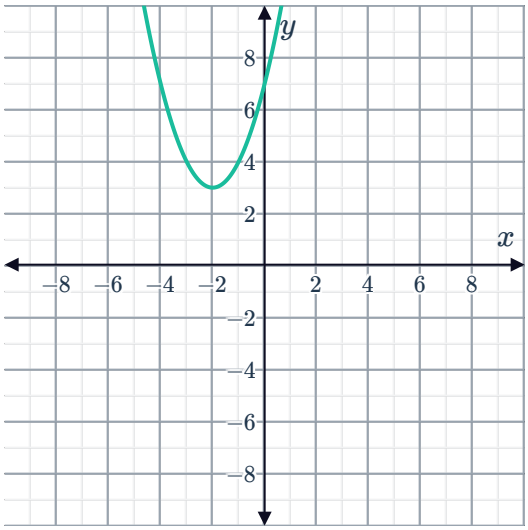
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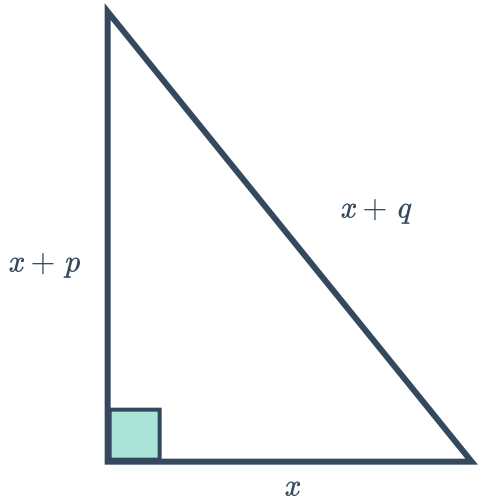
p



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Applications

31 Consider a right-angled triangle with side lengths x units, $x + p$ units and $x + q$ units, ordered from shortest to longest. No two sides of this triangle have the same length.



- **a** Complete the statement:
  p and q have lengths such that
  $0 < \square < \square$.
- **b** Write a quadratic equation that describes the relationship between the sides of the triangle in terms of x.
- **c** Find the discriminant of this quadratic equation.
- **d** How many real solutions are there to the quadratic equation?
- **e** Find the value of x when $q = 2p$. Give your answer in terms of p.

32 A cannon is fired from the top of a cliff 26 meters above sea level. The height of the cannon-ball above sea level, in meters, after t seconds have passed is given by the formula:

$$h = -5t^2 + 56t + 26$$

- **a** If we want to find out when the cannon-ball will reach a height of 200 meters, what equation do we need to solve?
- **b** Solve the equation found in part (a) for t.
- **c** Does the cannon-ball reach a height of 200 meters? Explain your answer.

33 An object is launched from a height of 90 feet with an initial velocity of 131 feet per second, and after x seconds its height (in feet) is given by $h = -16x^2 + 131x + 90$.
Solve for the number of seconds x after which the object is 20 feet above the ground.
Round your answer to the nearest tenth of a second.

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